

SIEMENS

MAGNETOM

MR

Software Information for Service

syngo MR D13 (VD13A)

MAGNETOM Avanto Dot / Verio Dot

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MAGNETOM

Software Release

NUMARIS/4 VD13A

syngo MR D13

N4_VD13A_LATEST_20120616_P1

N4_VD13A_LATEST_20120616

for the systems:

MAGNETOM Avanto Dot

MAGNETOM Verio Dot

July 9, 2012

Siemens AG
Healthcare Sector

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Release summary

Release Data

Software version: **syngo** MR D13

NUMARIS/4 VD13A

including the following packages:

syngo VX79B (SL07 Package 08)

COEM VE30A (COEM-VE30A01P01)

Argus VA80C

Vessel View VA30E

Tissue 4D VD13A

3DNeuro VA30D

AX Composer VB20B

syngo VH10A

syngo VH10A

Label: VH10A_SL77P112_RM01

COEM VZ10B_SL05P05

Ubuntu 10.04.1 LTS

MNP VA21B

CODIAG VA10A

MrFtk/ Autoalign Heart VA20A

Neuro Perfusion MoCo VA10A

AutoAlign Head (landmark) NUMARIS/4 VA30C

AutoAlign Knee H IM SY CAD VA10A

AutoAlign Spine H IM SY CAD VA10A

AutoAlign Shoulder H IM SY CAD VA10A

AutoAlign Hip H IM SY CAD VA10A

AutoAlign Breast H IM SY CAD VA10A

AutoAlign Aorta H IM SY CAD MR VA10A

AutoAlign Chest H IM SY CAD MR VA10A

AutoAlign Abdomen H IM SY CAD MR VA10A

AutoAlign Pelvis H IM SY CAD MR VA10A

Breast motion correction algorithm H IM SY CAD VA10C

MR Composer VA81

Adaptive Inline Filter VA10A

IQShim V4.1.1.5 Database 2.4.5

Products: MAGNETOM Aera, Skyra, Avanto Dot, Verio Dot

Software Label and Delivery Media

Software version: **syngo** MR D13

NUMARIS/4 VD13A

Software baseline: N4_VD13A_LATEST_20120616

N4_VD13A_LATEST_20120616_P1

The software baseline can be viewed with the function "Help, Info...":

COEM VZ10B_SL05P05
syngo VH10A_SL79P118
N4_VD13A_LATEST_20120616
N4_VD13A_LATEST_20120616_P1

The **syngo** MR D13 software has been developed according to valid software engineering and quality regulations. The software has undergone extensive testing through different test phases.

The contents of the released software are described in chapter 1 and as well as in chapter 2 of "Hints for Service".

Hardware and Operating Systems

Host (MR AWP and MR WP):

- Host Computer Fujitsu Technology Solutions Celsius M470 with FX 3800 graphic card
MR AWP 098 M470-2 (SNR 10660934)
MR-VD WP M470 DU (SNR 10746383)
- Host Computer Hewlett-Packard z400 with Q2000 graphics card
Grundrechner MR VD AWP Z400 (10746384)
Grundrechner MR VD WP Z400 (10746383)

Host operating system: Windows7

Imager

- DOT MARS
- Operating system Ubuntu-Linux with Kernel 2.6.35

Released Cameras

AGFA Drystar AXYS
AGFA Drystar 4500
AGFA Drystar 5300
AGFA Drystar 5302
AGFA Drystar 5500
AGFA Drystar 5503
CODONICS HorizonCODONICS
FUJI Drypix Plus
FUJI Drypix 7000 / 5000
FUJI Drypix 4000
FUJI Drypix 3000
FUJI Drypix 2000
FUJI Drypix 1000
FUJI FM DPL/FN-PS551
CARESTREAM DryView 5700
CARESTREAM DryView CHROMA
CARESTREAM DryView 6850
CARESTREAM DryView 5800
CARESTREAM DryView 5850
CARESTREAM DryView 6800
CARESTREAM DryView 8900
CARESTREAM DryView 8700-Plus / PACS Link MIM200
CARESTREAM DryView 8200 / PACS Link MIM 200
CARESTREAM DryView 8100 / PACS Link MIM 200
CARESTREAM DryView 8150
KONICA DryPro 873
KONICA DryPro 832
KONICA DryPro 793
KONICA DryPro 771
Konica DryPro 751/752
SONY UP-DF 750
SONY UP-DF 550
SONY UP-DF 500

Test phases

syngo MR D13 has completed the following test phases:

- Unit test (internally performed at SIEMENS by SW engineering teams)
- Integration test and validation of the **syngo** software (performed internally at SIEMENS by the SW engineering teams)
- System test (performed internally at SIEMENS by MR test teams and application specialist teams)

Service Features for **syngo** MR D13

syngo Remote Assist

- The “syngo Remote Assist” service offering allows remote desktop sharing of syngo-speaking MR systems via Siemens Remote Service infrastructure.
- Our goal is to provide qualified application and service support to our customers via remote desktop sharing.
- Moderators will be able to invite for syngo Remote Assist sessions. Several participants can be invited by the moderator and join the same session.
- Each remote participant access of the syngo MR desktop (screen) must be separately confirmed by the local syngo MR system operator.

SeSo General

PDF generator:

- allows generating PDF's from various information's available via the service software: "TuneUp", "Quality Assurance" and "TestTools" results, Start-up Report, Imprint, etc.

Eventlog:

- New layout with more filters and message Feedback function in the Message Description GUI.

Installation & Maintenance Protocol:

Start-up Report:

- Fully integrated in the system (service software)

Maintenance Protocol:

- Fully integrated in the system (service software)
- The maintenance protocol manages information e.g. when "Previous Maintenance" has been performed and when the "Next Maintenance" is to be performed.
- Safety related tests are integrated in the service software

Installing software

For a detailed description of how to install the System Software, refer to Installation Software for NUMARIS/4 VD13A.

System Image Backup

With the new function "**Image Backup**" the system will create Image a Backup of the C drive that it stores at the D. drive. The system image of the C: drive will be copied to the MARS service partition via the internal network as second copy.

Backing up site-specific data on external USB device

Backup & Restore at external USB devices is supported and tested.

Service

For the following service activities: (Configuration, Backup & Restore, TuneUp, Shim, Magnet & Cooling, Install & Maintenance, Quality Assurance, TestTools, DICOM Tests, File & Image Tools, Auto Report, and Utilities) the system provides information, see "Service UI" -> Help -> Online Help.

Host

Make sure to follow the instructions concerning important milestones during the software installation of the MR AWP / MR WP (MRC / MRSC):

- No user actions are allowed during the automatic installation. Wait until the login window is displayed.

DOT MARS

- The MARS software installation is triggered from the Host in the Service GUI > **TestTools > Trouble Reset > MCIR** > set the radio-button **Install** and click **Go**. The complete software of the MCIR (MARS) including the operating system is reinstalled. The MARS software is stored on the host (MRAWP).

Asset Management

Asset Management is important for the medical system product support care process. Due to regulatory obligations, any medical equipment sold by SIEMENS Healthcare Sector must be traceable for safety relevant updates/upgrades.

In this software version, only some of the replaceable components automatically provide their material number and serial number information to the system.

To see the "Asset Management Report" locally or remotely select :

Reports > Asset Management

Replacement of parts

After replacement of parts **a system reboot is required** to ensure that the asset update takes place; it changes "TuneUp" and "Quality Assurance" workflow steps from **Done** -> **ToDo**.

Configuration SRS / DICOM

Remote diagnostics

IE 8.0 has been tested and released as the service browser for remote service on **syngo** MR systems running NUMARIS VD13A.

The "SIEMENS Remote Services TM" will only support router connections for NEWLY delivered systems. See <https://intranet.healthcare.siemens.com/cms/CS/en/> "For Service", "SRS Toolkit" for the most recent information regarding the router configuration.

The project manager is responsible for planning a router connection to the Siemens Remote Diagnostic server. In case of questions concerning your pre-configured router from Siemens, please contact mailto:srs_final.med@siemens.com or your remote diagnostics person responsible.

CAN24 requires an own IP address in the customer LAN. The CAN24 must be in the same sub-network as the MRC (SRS Target and gateway is automatically cloned from the MRC).

Factory pre-configured Auto reports are available for all newly delivered NUMARIS/4 VD13A **syngo** MR systems. You only need to manually activate the pre-configured auto reports, if your customer agrees to remote service.

DICOM

The DICOM Standard is under continuous maintenance, the current official version is available at <http://dicom.nema.org>

The "CCC Team" provides a powerful configuration management tool.

Please refer to <https://intranet.healthcare.siemens.com/cms/CS/en/> "For Service", "DICOM", in the DICOM Homepage (CCC). Navigate to Databases "Interoperability DB", "Download", "Acquisition Tool".

Eventlog Feedback

Improving Event Message Quality through on-site Message Feedback:

- Message descriptions which leave room for improvement (e.g. hex-dec conversion)
- Unclear message instructions (e.g. component replacement suggested whereas component seems fine)
- Error messages which seem to occur without indicating actual errors

When you open an **Message Description** from the **EventLog Viewer** there is a **Feedback** button at the bottom right; this will provide a text input frame; after entering your "Feedback" click **Save**.

System

Software

Function	User administrator		
Problem	COBR hides the first disk to be able to boot then from the COBR DVD		
Workaround	For user administrator only US-English is supported		
	Version	syngo MR	Solution (planned) n.a.

Function	Mrvcheck		
Problem	Message: Findings occurred Check logfile C:\Medcom\log\MRV check log. is displayed to the customer if any additional SW or drivers has been installed.		
Workaround	Viewing the log content: Select System > Control select the "Tools" tab > select "View Consistency Log" How to Reset MRVCheck: In this case rename the file MrVcheck.ref to MrVcheck.ref_old under c:\medcom\config\mri\mrsystem A new reference file with the actual state will be created during the next re-boot		
396452	Version	syngo MR	Solution (planned) n.a.

Service

Backup & Restore

Function	Backup site specific data on CD		
Problem	Backup on CD is not possible		
Workaround	Backup the data on the c drive; use the Windows 7 CD burn function		
	Version	syngo MR	Solution (planned) n.a.

Function	Restore site specific data		
Problem	First boot after restore of site specific data syngo MR applications like Archnet, Image transfer, etc might not work due to missing synchronization in between network configuration setting and the windows operating system.		
Workaround	Re-boot the system.		
	Version	syngo MR	Solution (planned) n.a.

Function	Restore site specific data		
Problem	The Image Text settings are not part of backup/restore from VB17A to VD13A		
Workaround	Configure the Image Text settings manually		
	Version	syngo MR	Solution (planned) n.a.

Function	Restore site specific data		
Problem	Postscript (paer) Printer which are configured via the Windows configuratin are generally not part of backup/restore.		
Workaround	Configure the printer settings manually		
	Version	syngo MR	Solution (planned) n.a.

Function	Restore site specific data: Meas. Settings		
Problem	TheMeas. Settings are not automatically activated.		
Workaround	- Log in to the service platform and select Configuration. - In the menu Meas. Settings click Save > Finish > Home - Confirm the system reboot.		
	Version	syngo MR D13	Solution (planned) n.a.

Configuration

Function	Change of meduser password		
Problem	If the password is changed for meduser, the connection to the measurement system might fail.		
Workaround	Do not change the password for meduser / set back the meduser default password.		
	Version	syngo MR	Solution (planned) n.a.

Function	IPv6		
Problem	IPv6 is not supported for Numaris VD13A		
Workaround	Only use IPv4		
	Version	syngo MR	Solution (planned) n.a.

Function	Date of birth, Japan		
Problem	The Japanese date format (called "Wareki" or "Nengo") is not supported by the syngo interface. The traditional Japanese calendar consists of eras based on the reign of a particular emperor (e.g., the year 2004 is the 16th year of the Heisei era which began in 1989 as year 1).		
Workaround	Imperial dates are formatted with the name of the era followed by year, month, and day. Set date and time to yyyy-MM-dd as follows: Options ->, Configuration, double click Regional and Language Options. Click Customize, click Tab date, select yyyy-MM-dd in the pull-down menu "Short date format" Enter: yyyy-MM-dd in the input field "Long date format"		
	Version	syngo MR	Solution (planned) n.a.

Function	Configuration of system language		
Problem	If advanced user is enabled during configuration of system language, the system will not re-boot automatically.		
Workaround	Re-boot the system manually.		
	Version	syngo MR	Solution (planned) n.a.

Function	Configuration of printer		
Problem	After changes in the printer configuration, the system boot displayed one pop-up with a fatal error message.		
Workaround	Acknowledge the error message.		
	Version	syngo MR	Solution (planned) n.a.

TestTools

Function	TestTools concept change		
Problem	As the test concept of the VA/VB/VC-software was changed with VDx, also the transferred tests had to be re-organized. The tests are no longer executed by selecting individual components or FRUs, but by choosing functional units. For example: "TX" includes components which create, transmit, and amplify RF power signals (RFPA, TAS, TALES, BC); "RX" contains the components that processes the small amplitude signals (transmitter, receiver, RCCS). Here the functions of the BCCS are split: the power-part is contained in the TX category, the receive signal part is in the RX category. As far as possible, redundant measurements are omitted by applying all necessary evaluations on a single data set. For the RFPA for example, the maximum power, the linearity as well as the reflection of RF power are checked simultaneously.		
Workaround	na		
	Version	syngo MR	Solution (planned) n.a.

Function	Test Cables		
Problem	For some interactive Control tests within the TestTools of Avanto and Verio dot additional test cables are necessary.		
Workaround	They can be ordered as a set by use of the part number 10759471 • Cables for the LWL_INT MDSD Interface and dV_Star RS232-Loop test: One/two plastic Fiber Optic Cables • Cables for the CAN Connector test: Two 9-pin SUB-D connectors The remaining cables of the set are for general test purposes and troubleshooting to figure out if a cable is defective or the connected components. They can be used for Aera and Skyra, too.		
	Version	syngo MR	Solution (planned) n.a.

Function	Service Plug Mode		
Problem	The handling of the Service Plug used for Avanto and Verio dot systems beginning with the VD-software differs from the way how the Service Plug is used in previous software versions. There for Avanto and Verio the mode is always "TX", whereas with the VD-software this depends on the test that is performed. This means that tests with both the TX and the RX mode are present. For TX tests the Service Plug has to be set to "TX", for RX tests like the LC Plug tests the mode has to be "RX".		
Workaround	The popups of the tests instruct the user which mode has to be used. If that is not followed the test will fail or the system doesn't start the workflow at all (see also next item "error handling"). Please note that with some LC Plug tests and specific sockets also the TX mode would be possible. But don't do that to get used to the new handling of the Service Plug! At these sockets the mode of the Service Plug has no influence and the test is performed regularly. This is not an error and is based on the behaviour of the measurement system.		
	Version	syngo MR	Solution (planned) n.a.

Function	Multi Nucleus Option		
Problem	The interactive TX LC tests at Verio dot systems with installed Multi Nucleus Option (MNO/MKO) behaves different than you might expect. There for the LC TX path by default always RFPA2 is used at coil socket 1, never RFPA1. This is not an error and is based on the behaviour of the measurement system. It is not possible to switch over to RFPA1 for test purposes. The result is that the standard interactive TX LC test also uses RFPA2 and not the probably expected RFPA1. With Avanto dot systems the behaviour is as expected as the RFPA2 is not able to transmit on the 1H frequency, whereas the RFPA2 at the Verio system is able to do so.		
Workaround	na		
	Version	syngo MR	Solution (planned) n.a.

Function	Error handling for tests using the service plug		
Problem	The error handling of the Avanto and Verio dot tests that make use of the Service Plug has some gaps when checking the operation mode of the Service Plug. Some tests do not detect/check the correct mode of the Service Plug. This will be fixed with a later software version. • Interactive TX LC - A missing Service Plug is detected and a corresponding error message displayed. - A wrong setting of the Service Plug ("RX" instead of "TX") is not detected, the test is started and fails at the end with "out of specification". • Interactive LC Plug 1 RX and LC Plug 1 PIN - A missing Service Plug is detected and a corresponding error message displayed. - A wrong setting is not detected but the test is performed successfully with the same results both with the mode set to "TX" and "RX". - On systems equipped with the Mult Nucleus option (MNO) this behaviour can be observed also with coil socket 2. • Interactive LC Plug 2 - 10 RX and LC Plug 2 - 10 PIN - A missing Service Plug is detected and a corresponding error message displayed. - A wrong setting is detected already while plugging in. The message "#<socket number> wrong socket" ("#2 ..." through "#10 ...") is displayed at the magnet display. It is not possible to start the test as this is blocked by the measurement system and thus also by the test program. - On systems equipped with MNO this behaviour starts with coil socket 3. Here coil socket 2 also allows the use of LC-TX coils. The reason is that LC-TX coils are not permitted at coil socket 2 to 10 at normal systems and 3 to 10 at systems with MNO. • Multi Nucleus TX - A missing Service Plug is detected and a corresponding error message displayed. - A wrong setting of the Service Plug ("TX" instead of "RX") is not detected, the test is started and fails with a measurement system error. • Multi Nucleus TX LC - A missing Service Plug is detected and a corresponding error message displayed. - A wrong setting of the Service Plug ("TX" instead of "RX") is not detected, the test is started for 1H only and fails at the end. - With the correct setting the test is performed for all available nuclei, not only for 1H!		
Workaround	n.a.		
	Version	syngo MR	Solution (planned) n.a.

Function	Error messages interactive Control tests
Problem	In case one or more of the the interactive Control tests of Avanto and Verio dot fail, no diagnose texts are printed out at the end of the test report. The bug will be fixed with a later software version.
Workaround	Please use the following tests instead: LWL_INT - AMC_IO Transceiver test Possible error sources: • LWL_INT D6; • AMC_IO D20; • AMC backplane D5. Please proceed as follows as long as the error persists: • re-establish the original cabling, reboot "Meas&Recon", and repeat the test; • if the error still persists, replace one of the components listed above, repeat the test, and follow up accordingly. LWL_INT MDSD Interface test Possible error sources if the test fails: • LWL_INT D6; • AMC_IO D20; • AMC backplane D5. Possible error sources if the test doesn't fail but there are still problems with the PTAB: • Fiber Optic Cable (FOC) W4118 between PTAB/MDSD A4320/4330 U7/U8 and LWL_INT D6 U4/U5; • Horizontal Drive MDSD A4320/4330 (Encoder). Please proceed as follows as long as the error persists: • re-establish the original cabling, reboot "Meas&Recon", and repeat the test; • if the error still persists, replace one of the components listed above, repeat the test, and follow up accordingly.

Workaround	CAN Connector test Possible error sources: • cable between dV_Star D4 X22 and Sig_distr2 D23 X1; • cable between Sig_distr2 D23 X2 and CAN_Connectors D21 X5; • dV_Star D4; • Sig_distr2 D23; • CAN_Connectors D21; • AMC backplane D5. Please proceed as follows as long as the error persists: • check the cabling between the dV_Star and the Sig_distr2 boards; • check the cabling between the Sig_distr2 and the CAN_Connectors boards; • check the dV_Star and the Sig_distr2 boards in the MARS for correct seating; • check the CAN_Connectors board in the AMC for correct seating; • if the error persists re-establish the original cabling, reboot "Meas&Recon", and repeat the test; • if the error still persists, replace one of the components listed above, repeat the test, and follow up accordingly. dV_Star RS232-Loop test Possible error sources: • cable between dV_Star D4 X22 and Sig_distr2 D23 X1; • cable between Sig_distr2 D23 X1 and RS232_Adapter D22 X1; • dV_Star D4; • Sig_Distri2 D23; • RS232_Adapter D22; • LWL_INT D6; • AMC backplane D5. Additional possible error sources for missing PMU beeps: • cable between AMC backplane X25 and ACC roof X88; • cable between ACC roof X88 and Central Intercom Unit X10; • cable between Central Intercom Unit X12 and Operating Unit Intercom; • Central Intercom Unit; • Operating Unit Intercom.
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Workaround	Please proceed as follows as long as the error persists: • check the cabling between dV_Star and the Sig_distr2 boards; • check the cabling between the Sig_distr2 and the RS232_Adapter boards; • check the dV_Star and the Sig_distr2 boards in the MARS for correct seating; • check the RS232_Adapter and the LWL_INT boards in the AMC for correct seating; • if the error persists re-establish the original cabling, reboot "Meas&Recon", and repeat the test; • if the error still persists, replace one of the components listed above, repeat the test, and follow up accordingly. In case no PMU beep was audible please proceed additionally as follows: • check the functionality of the intercom (microphone, loudspeaker); • perform the PMU test in the test category Interactive and follow up accordingly; • if the error still persists, check the cabling between AMC backplane X25 and ACC roof X88; • check the cabling between ACC roof X88 and Central Intercom Unit X10; • check the cabling between Central Intercom Unit X12 and Operating Unit Intercom; • if the error still persists, replace one of the components listed above, repeat the test, and follow up accordingly.		
	Version	syngo MR	Solution (planned) n.a.

Filming

Function	UI crashes with Runtime error		
Problem	Following error is displayed: Exception caught while creating filming config objects Component FilmingServer initialization failed. Info: Filming BE is shutting down		
Workaround	Run script: %MEDHOME%\utils\Init_MR_Layouts_V3.bat Follow the instructions provided by the script.		
387640	Version	syngo MR	Solution (planned) n.a.

Initial version

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